direction. (For further discussion of the Lennard-Jones few-body problem from a dynamical systems perspective, the reader is referred to [87].) □

1.7.1 Global Solutions

Looking at the trajectories of a molecular dynamics simulation involving even a few atoms (such as the restricted trimer model, above) suggests their erratic, irregular aspect. Individual paths will not be expected to be predictable in any practical sense—every time one atom collides with another it will exchange a small amount of energy and momentum with the other atom, the precise amounts depending in an intricate way on the motions of all the atoms of the system, and the atoms then move on in perturbed directions. In a very short time, the state of the system is well stirred by the molecular interactions. This is one of the elements of the common view of what constitutes a chaotic dynamical system.

In the words of H. Poincaré, writing over a century ago [304],

> A very small, unnoticeable cause can determine a visible very large effect; in this case we claim that this effect is the product of random chance. However, even if the natural laws were perfectly known, we would only be able to know the initial conditions with some approximation. If this allows us to know the future with the same approximation that is all we need. We will say that the phenomenon is foreseeable, that it is governed by laws. However this is not always the case, it is possible that a very small initial difference leads to a very large change in the final state...

Poincaré's comments probably derived from his understanding of the three-body gravitational problem, but they would apply as well to a typical molecular system.

Typical definitions of a "chaotic dynamical system" [103] require at least the following conditions to be satisfied on the phase space $\mathcal{D}$:

- The solutions depend sensitively on the initial data taken from $\mathcal{D}$;
- The flow is topologically transitive in $\mathcal{D}$.

Let us address these in turn, without being entirely formal. The sensitive dependence on initial conditions can be taken to mean that if a pair of initial points of phase space is given which are separated by any finite amount, no matter how small, then the gap between these solutions grows rapidly (typically exponentially fast) in time. A problem with this concept is that we often think of molecular systems as having an evolution that is bounded by some sort of domain restriction or a property of the energy function: the exponential growth for a finite perturbation can therefore only be valid until the separation approaches the limits of the accessible region of phase space. In order to be able to make sense of the calculation of an exponential rate in the asymptotic ($t \to \infty$) sense, we need to consider infinitesimal perturbations of the initial conditions, and this can be made precise by consideration of the Lyapunov characteristic exponents mentioned at the end of this chapter.

The second condition states that given arbitrarily small neighborhoods $\mathcal{D}_1$ and $\mathcal{D}_2$ of two different points in $\mathcal{D}$ then it is possible to find a trajectory that goes

from some point of $\mathcal{D}_1$ to some point of $\mathcal{D}_2$. This condition roughly says that the trajectories starting from some arbitrarily small open set in $\mathcal{D}$ will eventually fill the interior of $\mathcal{D}$ densely. We will see later that this concept, which is essentially equivalent to *ergodicity*, is a crucial component of molecular theories. Below, we will demonstrate, using numerical experiments, that even a simple nonlinear system reminiscent of molecular dynamics can exhibit the hallmarks of a chaotic system. After a short time, small perturbations of the initial conditions will become large, of order of the dimension of the system, and the "memory" of the initial conditions is lost. In this sense the arrangement of atoms rapidly becomes disordered in a similar way to the playing cards of a deck as they are shuffled. The idea of a chaotic dynamical system and its interpretation as a randomizing process is further developed in later chapters. For an informal but valuable introduction to this topic the reader is referred to the short book of Ruelle [318].

Example 1.9[7] *(Anisotropic Oscillator)* A perturbation of the central force problem illustrates the consequences of a breakdown of integrability. Consider the system with energy

$$E(x, y, \dot{x}, \dot{y}) = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) + \frac{\kappa(c_3)}{2}(r - l(c_3))^2, \qquad r = \sqrt{x^2 + y^2}, \tag{1.9}$$

where

$$c_3 = \cos(3\theta)$$

is defined in terms of the angular coordinate of the position (x, y) with respect to the $(1, 0)$-direction,

$$\cos\theta = c = \frac{x}{r}, \qquad c_3 = 4c^3 - 3c,$$

and we have defined

$$\kappa(c_3) = \kappa_0(1 - \frac{1}{2}\varepsilon c_3), \qquad l(c_3) = l_0(1 + \frac{1}{2}\varepsilon c_3).$$

Effectively this models the mythical behavior of an atom that binds more tightly to a molecule in certain configurations than in others. In our experiments we took, for simplicity, $l_0 = \kappa_0 = 1$, and considered the effect of varying ε. For $\varepsilon \neq 0$, the angular momentum is not preserved, and the only invariant is the energy.

When ε is small one finds trajectories have a regular appearance for many values of the initial conditions, typically either a periodic orbit or one which repeatedly and *quasiperiodically* traverses a portion of the domain. As ε is increased, the orbits

[7] The Anisotropic Oscillator is included as an example in the **MD.M** package, see http://MolecularDynamics.info